

Chapter 2: Gerber and Green — Causal Inference and Experimentation – Maria Capi

Exercise 2.1: Potential Outcomes Notation

A. (Y_i^0) is the potential outcome for unit (i) if that unit does not receive the treatment.

B. The untreated potential outcome for a unit that was assigned to the treatment group. In other words, it is what would have happened to that unit if it had not received the treatment, even though it was assigned to be treated.

C. The untreated potential outcome for any unit VS. the untreated potential outcome specifically for units assigned to the control group.

D. The first is the untreated potential outcome for units assigned to treatment. The second is the untreated potential outcome for units assigned to control. The difference shows whether the two groups would have had different outcomes even without treatment.

E. The average untreated potential outcome for the whole population. And the average untreated potential outcome only for those assigned to treatment.

F. Why is selection bias equal to zero?

With random assignment, the treatment and control groups should be similar on average. Their untreated potential outcomes should therefore also be the same on average. This makes selection bias equal to zero:

Exercise 2.10: Free Newspaper Experiment

A. Excludability

The excludability assumption means that assignment to the treatment group should affect students' political interest only through receiving the free newspaper.

The disclosure letter may violate this assumption because it tells students that they are taking part in a study about political interest. The letter itself may make them think more about politics or change how they answer the questions, even before they receive or read the newspaper.

A student who receives the disclosure letter may respond differently from a control student, even if neither student reads the newspaper.

B. Non-interference

The non-interference assumption means that one student's outcome should not be affected by the treatment assignment of another student.

This assumption may be violated because treated students could bring their newspapers into shared spaces, such as cafeterias or dormitories. Control students may read the newspapers or take part in

political conversations started by treated students. Their interest in politics could therefore increase even though they were assigned to the control group.

Exercise 2.12

A. Observational study

The prisoners self-select into the two groups: some naturally read for at least three hours and some do not. The researcher is therefore finding a correlation, not necessarily a causal effect.

The groups may already be different. For example, calmer prisoners may be more likely to read and also less likely to have violent encounters. Therefore:

$$E[Y_i(0) \mid D_i = 1] \neq E[Y_i(0) \mid D_i = 0].$$

The difference may be caused by temperament or general behaviour, rather than by reading.

B. Excludability

The excludability assumption means that assignment should affect violent encounters only through reading.

This may not hold because the treatment includes more than reading. Prisoners are removed from their usual routine, placed in separate carrels, possibly supervised more closely, and given fewer opportunities for confrontation. The same effect might occur with another quiet activity, such as knitting. Therefore, it may be the isolation and reduced interaction, rather than reading itself, that reduces violence.

C. Non-interference

The non-interference assumption means that one prisoner's outcome should not be affected by the treatment assignment of other prisoners.

This may not hold because prisoners interact. They may discuss the programme, tease those assigned to read, or influence each other's behaviour. Removing some prisoners from normal activities may also affect the behaviour of the control group.

D. Applying the policy to all prisoners

The experiment tests what happens when only ten prisoners are required to read. If all prisoners were required to read, the whole prison environment could change.

There might be fewer opportunities for conflict, different group behaviour, changes in staff supervision, or overcrowding in the library. Therefore, the effect found for ten treated prisoners may not be the same when the policy is applied to everyone.